



Indian Institute of Information Technology, Surat

भारतीय सूचना प्रौद्योगिकी संस्थान, सूरत
(Institute of National Importance under Act of Parliament)

EC 503:

Image Processing and Computer Vision

IMAGE RESTORATION

By:
Dr. Hemant S. Goklani
ECE Department
I.I.I.T, Surat

IMAGE RESTORATION

- ❑ The ultimate goal is to improve an image in some **predefined** sense.
- ❑ Restoration
 - The given image is degraded.
 - Degradation is modeled.
 - Inverse process is applied.
- ❑ No focus on sensor, digitizer, and display degradations.
- ❑ Some restoration techniques are best formulated in the spatial domain while others are better suited for the frequency domain.
 - **Spatial domain: additive noise.**
 - **Frequency domain: blurring**

A MODEL OF IMAGE DEGRADATION/RESTORATION PROCESS

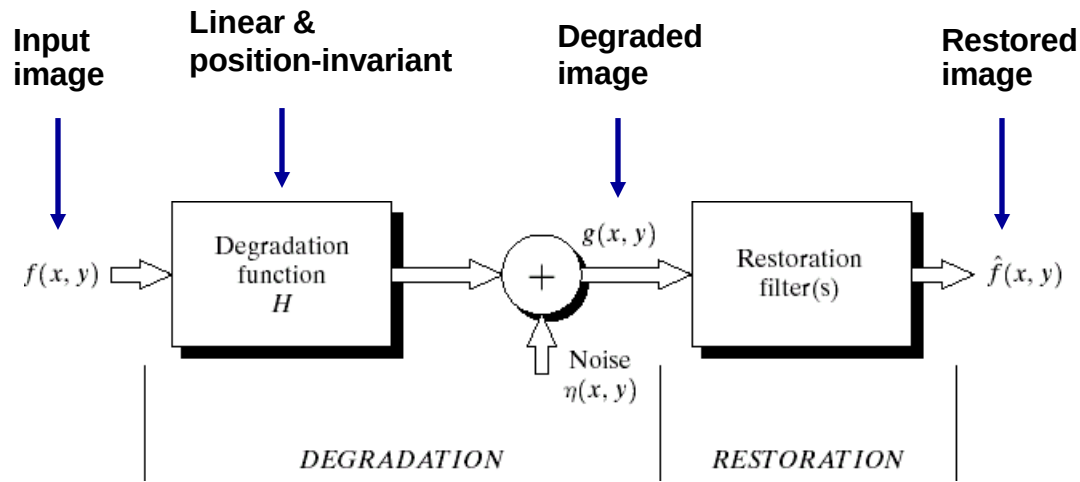


FIGURE 5.1 A model of the image degradation/restoration process.

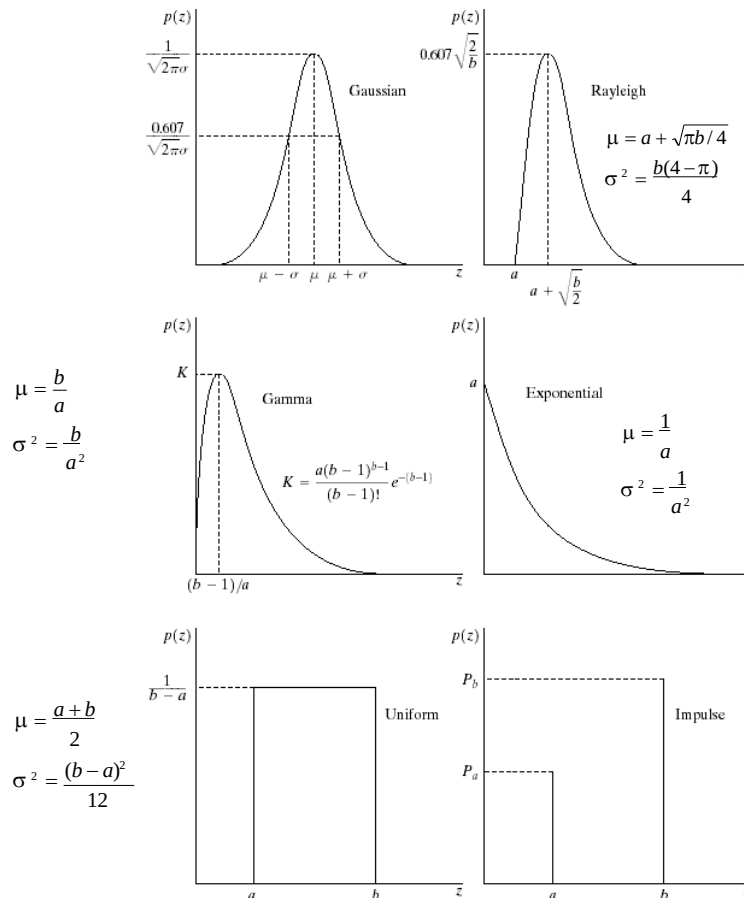
$$g(x, y) = h(x, y) * f(x, y) + \eta(x, y)$$

$$G(u, v) = H(u, v)F(u, v) + N(u, v)$$

NOISE MODELS

- ❑ **Principle sources of noise in digital images arise during acquisition and transmission.**
 - Acquisition: performance of image sensors is affected by a variety of factors (environmental conditions, etc.) and by the quality of the sensing elements.
 - Transmission: images are corrupted due to interference in the channel.
- ❑ **Spatial properties of noise**
- ❑ **Frequency properties of noise**
 - White noise: Fourier spectrum of noise is **constant**.
- ❑ **Assumption: noise is independent of spatial coordinates.**
 - Partially invalid.
 - No correlation between pixel values and noise.
 - Exception: spatially periodic noise.

NOISE PROBABILITY DENSITY FUNCTIONS



We are concerned with the **statistical behavior** of the gray-level values of the noise component of the model.

These values may be considered random variables characterized by a **probability density function (PDF)**.

Most common PDFs:

1. Gaussian (normal) noise
2. Rayleigh noise
3. Erlang (Gamma) noise
4. Exponential noise
5. Uniform noise
6. Impulse (salt-and-pepper) noise

MODELING A BROAD RANGE OF NOISE CORRUPTIONS

- ❑ The 6 PDFs are useful tools.
- ❑ **Gaussian noise** arises in an image due to factors such as electronic circuit noise and sensor noise due to poor illumination and/or high temperature.
- ❑ **Rayleigh density** is helpful in characterizing noise phenomena in range imaging.
- ❑ **Gamma and exponential densities** find application in laser imaging.
- ❑ **Impulse noise** is found in situations where quick transients, such as faulty switching, take place during imaging.
- ❑ **Uniform density** may be the least descriptive of practical situations but it is quite useful as the basis for numerous random number generators.

A TEST PATTERN

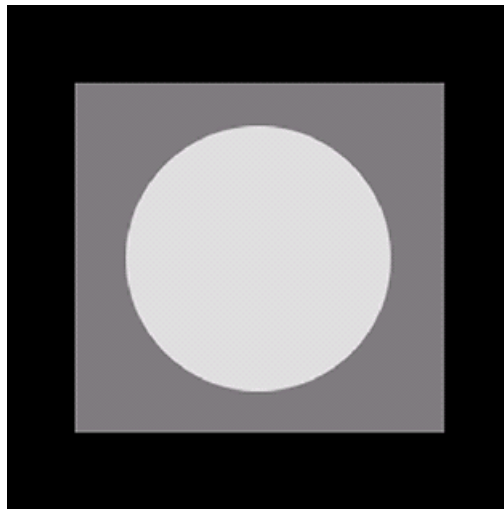
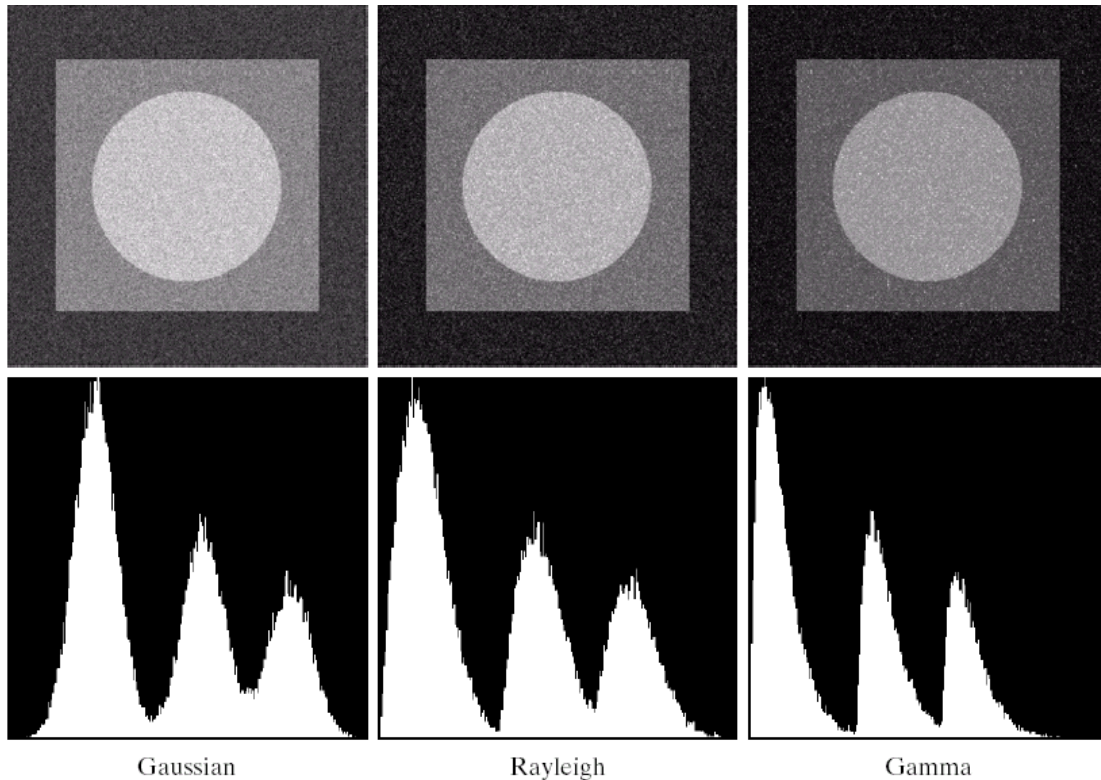


FIGURE 5.3 Test pattern used to illustrate the characteristics of the noise PDFs shown in Fig. 5.2.

composed of simple, constant areas that span the gray scale from **black** to **near white** in 3 increments.

NOISY IMAGES AND THEIR HISTOGRAMS – GAUSSIAN, RAYLEIGH, AND GAMMA

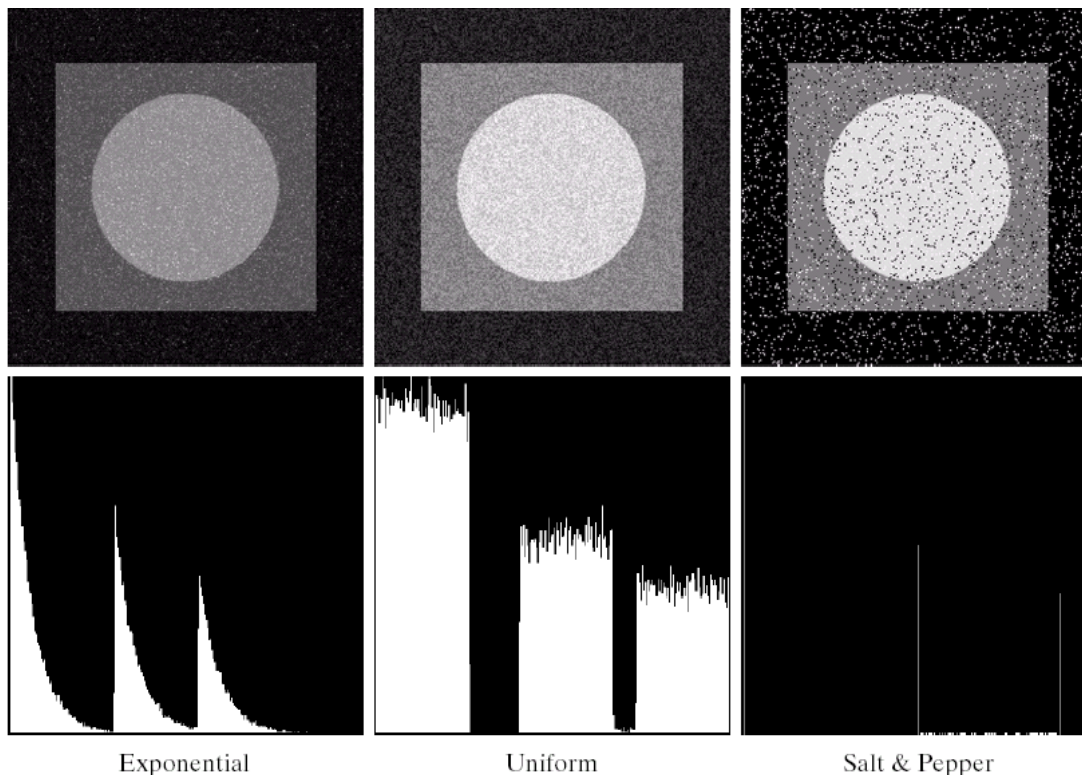


The parameters of the noise were chosen in each case so that the histogram corresponding to the 3 gray levels in the test pattern would start to merge.

a b c
d e f

FIGURE 5.4 Images and histograms resulting from adding Gaussian, Rayleigh, and gamma noise to the image in Fig. 5.3.

NOISY IMAGES AND THEIR HISTOGRAMS – EXPONENTIAL, UNIFORM, AND IMPULSE



The parameters of the noise were chosen in each case so that the histogram corresponding to the 3 gray levels in the test pattern would start to merge.

Extra peak: the noise components were pure black and white, and the lightest component of the test pattern is light gray.

g h i
j k l

FIGURE 5.4 (Continued) Images and histograms resulting from adding exponential, uniform, and impulse noise to the image in Fig. 5.3.

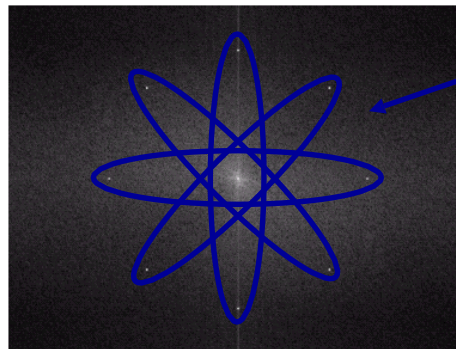
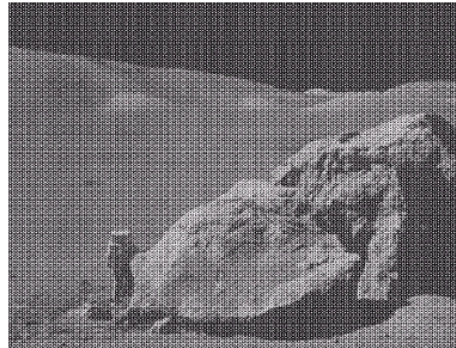
PERIODIC NOISE

a
b

FIGURE 5.5
(a) Image corrupted by sinusoidal noise.
(b) Spectrum (each pair of conjugate impulses corresponds to one sine wave).
(Original image courtesy of NASA.)

Periodic noise arises typically from electrical or electromechanical interference during acquisition.

This is the only **spatially dependent** noise considered here.



← Image is severely corrupted by **spatial sinusoidal noise** of various frequencies.

The Fourier Transform of a pure sinusoid is **a pair of conjugate impulses** located at the conjugate frequencies of the sine wave.

← If the amplitude of a sine wave in the spatial domain is strong enough, we would see in the spectrum of the image **a pair of impulses for each sine wave** in the image.

ESTIMATION OF NOISE PARAMETERS

❑ Periodic noise

- Inspection of the Fourier spectrum
- Inspection of the image (possible only in simple cases)
- Automated analysis
 - Noise spikes are exceptionally pronounced.
 - Some knowledge is available about the general location of the frequency components.

❑ Noisy PDFs

- Parameters may be partially known from sensor specs
- Imaging system available
 - Capture a set of images of flat environments.
- Images are available
 - Crop small patches of reasonably constant gray level.
 - Obtain the histogram.
 - Compute mean and variance.
 - Gaussian PDF: Completely determined by the mean and variance.
 - Impulse noise: actual probability of occurrence of white and black pixels is needed.
 - Others: Use the mean and variance to solve for a and b .

SPATIAL DOMAIN FILTERING FOR ADDITIVE NOISE

$$g(x, y) = f(x, y) + \eta(x, y) \quad \leftarrow \text{additive noise}$$

$$G(u, v) = F(u, v) + N(u, v)$$

Mean filters

- Arithmetic mean filter
- Geometric mean filter
- Harmonic mean filter
- Contraharmonic mean filter

Order-statistics filters

- Median filter
- Max & min filters
- Midpoint filter
- Alpha-trimmed mean filter

Adaptive filters

- Adaptive local noise reduction filter
- Adaptive median filter

MEAN FILTERS

$$\hat{f}(x, y) = \frac{1}{mn} \sum_{(s,t) \in S_{xy}} g(s, t)$$

Arithmetic mean

$$\hat{f}(x, y) = \left[\prod_{(s,t) \in S_{xy}} g(s, t) \right]^{\frac{1}{mn}}$$

Geometric mean
(comparable to arithmetic mean but tends to lose less image detail)

$$\hat{f}(x, y) = \frac{mn}{\sum_{(s,t) \in S_{xy}} \frac{1}{g(s, t)}}$$

Harmonic mean
(works well for salt noise but fails for pepper noise. OK for other types of noise as well)

Q=0 $\cup$ arithmetic mean

Q=-1 $\cup$ harmonic mean

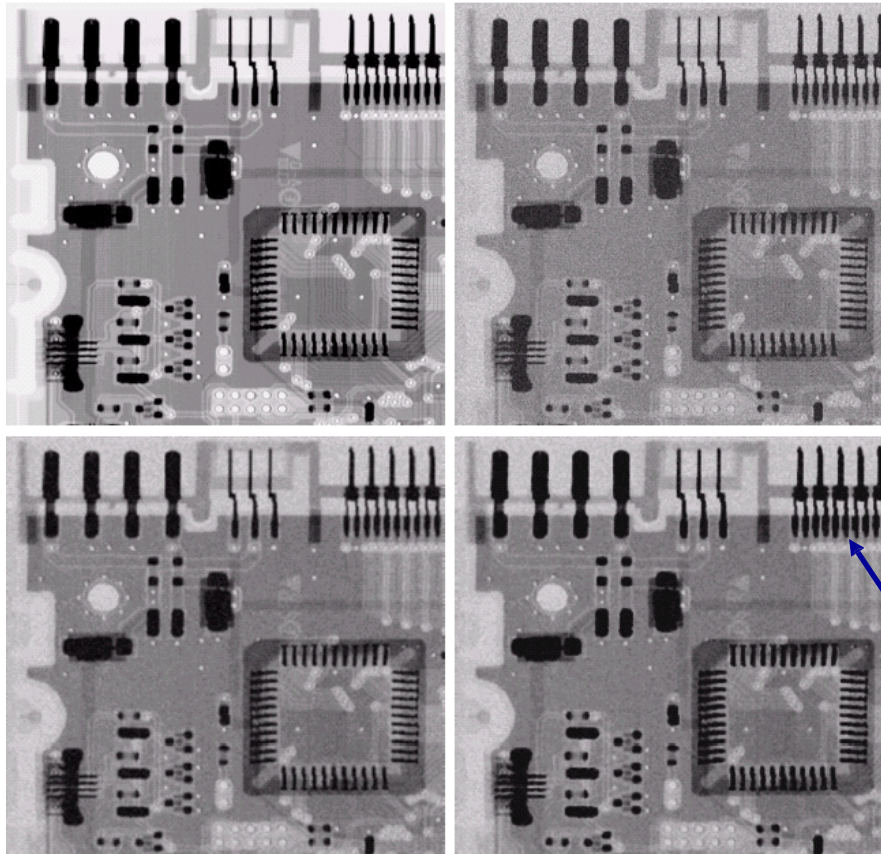
$$\hat{f}(x, y) = \frac{\sum_{(s,t) \in S_{xy}} g(s, t)^{Q+1}}{\sum_{(s,t) \in S_{xy}} g(s, t)^Q}$$

Order of the filter

Contraharmonic mean
(+Q: eliminates pepper noise
-Q: eliminates salt noise
not simultaneously!)

ARITHMETIC AND GEOMETRIC MEAN FILTERS

mean=0, variance=400

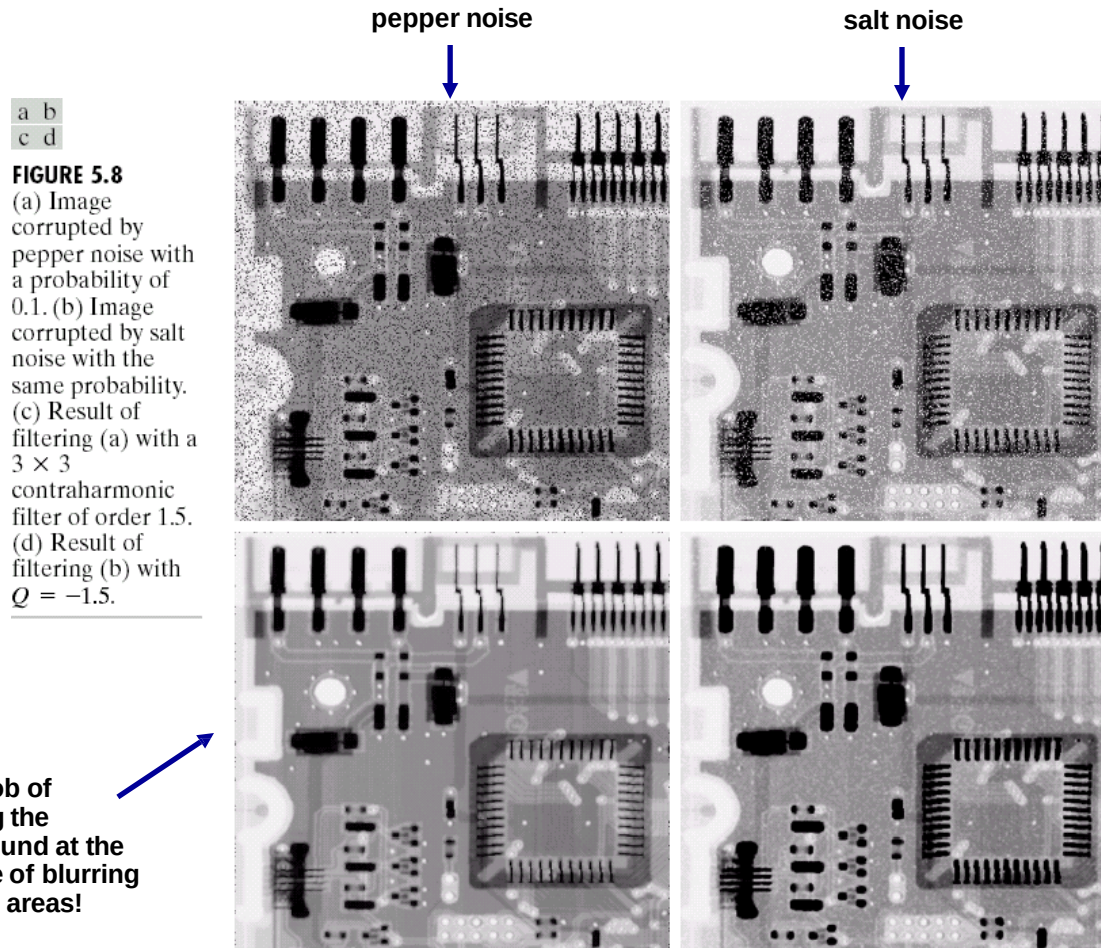


a	b
c	d

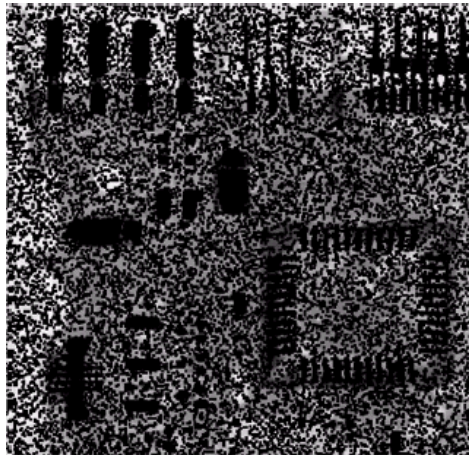
FIGURE 5.7 (a) X-ray image. (b) Image corrupted by additive Gaussian noise. (c) Result of filtering with an arithmetic mean filter of size 3×3 . (d) Result of filtering with a geometric mean filter of the same size. (Original image courtesy of Mr. Joseph E. Pascente, Lixi, Inc.)

Geometric mean filter did not blur the image as much as the arithmetic mean filter

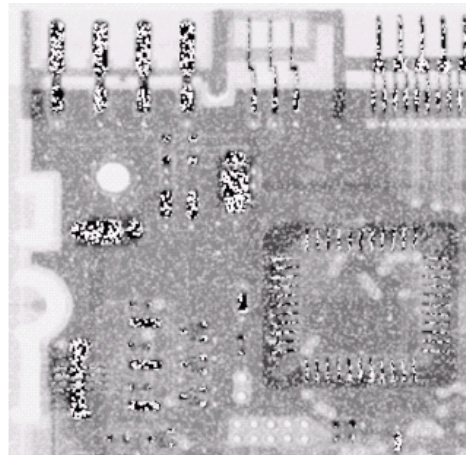
SPATIAL FILTERING FOR ADDITIVE NOISE



CONTRAHARMONIC FILTERING WITH THE WRONG SIGN



↑
Wrong sign for Q



↑
Wrong sign for Q

a b

FIGURE 5.9 Results of selecting the wrong sign in contraharmonic filtering. (a) Result of filtering Fig. 5.8(a) with a contraharmonic filter of size 3×3 and $Q = -1.5$. (b) Result of filtering 5.8(b) with $Q = 1.5$.

ORDER-STATISTICS FILTERS

Effective for
bipolar and
unipolar impulse
noise

$$\hat{f}(x, y) = \text{median}_{(s,t) \in S_{xy}} \{g(s, t)\}$$

Median filter

Useful for finding
the brightest
points in an image

$$\hat{f}(x, y) = \max_{(s,t) \in S_{xy}} \{g(s, t)\}$$

Max filter

Useful for finding
the darkest points
in an image

$$\hat{f}(x, y) = \min_{(s,t) \in S_{xy}} \{g(s, t)\}$$

Min filter

Works best for
randomly
distributed noise

$$\hat{f}(x, y) = \frac{1}{2} \left[\max_{(s,t) \in S_{xy}} \{g(s, t)\} + \min_{(s,t) \in S_{xy}} \{g(s, t)\} \right]$$

Mid-point filter

$d = 0$ $\Rightarrow$ arithmetic
 $d = (mn-1)/2$ $\Rightarrow$ median
Other d : Useful for
multiple types of noise

$$\hat{f}(x, y) = \frac{1}{mn - d} \sum_{(s,t) \in S_{xy}} g_r(s, t)$$

Alpha-trimmed filter

3 PASSES OF MEDIAN FILTER FOR IMPULSE NOISE

a b
c d

FIGURE 5.10

(a) Image corrupted by salt-and-pepper noise with probabilities $P_a = P_b = 0.1$.

(b) Result of one pass with a median filter of size 3×3 .

(c) Result of processing (b) with this filter.

(d) Result of processing (c) with the same filter.

2nd pass

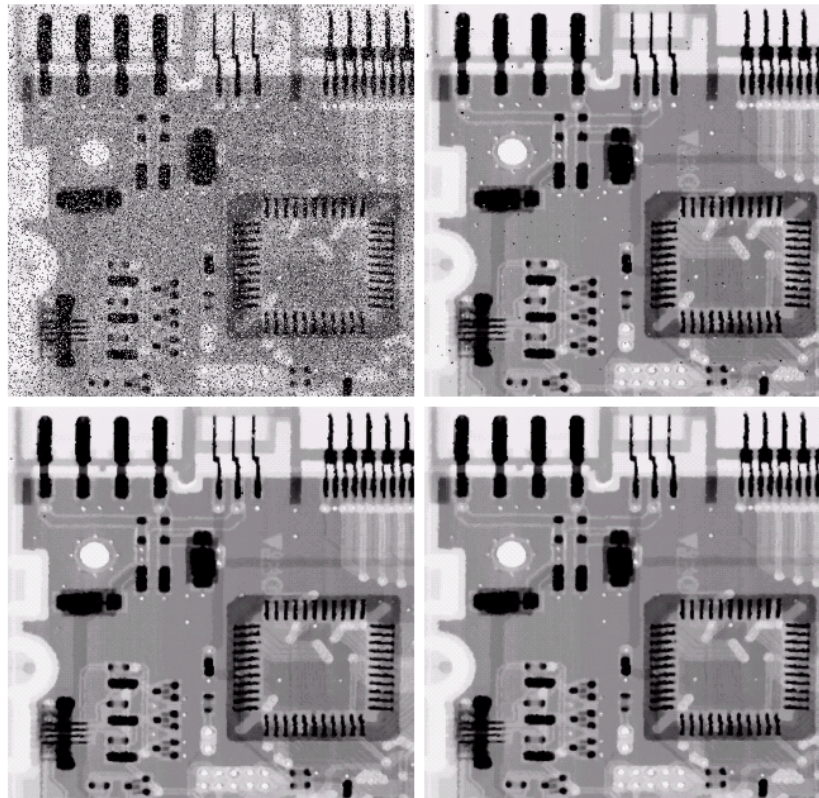
**barely visible
noise points!**

1st pass

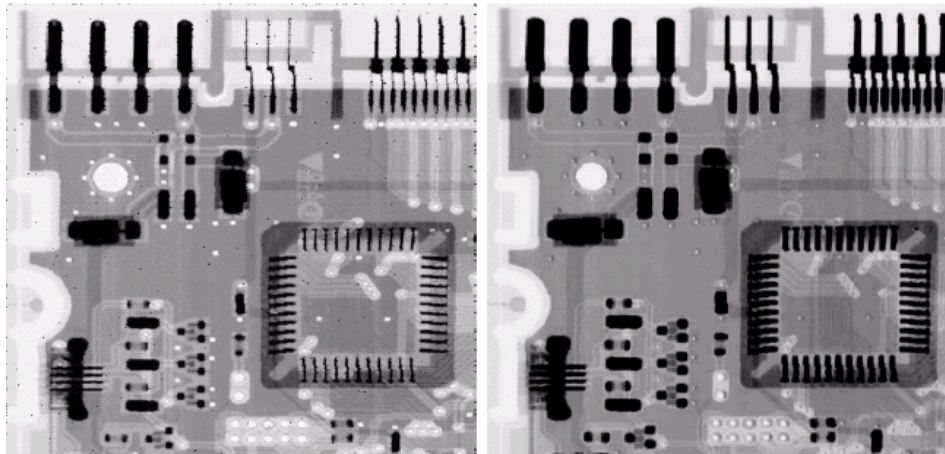
**Significant
Improvement!**

3rd pass

**# of passes
should be as
low as
possible!**



MAX & MIN FILTERS FOR PEPPER NOISE



a b

FIGURE 5.11
(a) Result of filtering Fig. 5.8(a) with a max filter of size 3×3 . (b) Result of filtering 5.8(b) with a min filter of the same size.

↑
Pepper noise is reasonably removed.

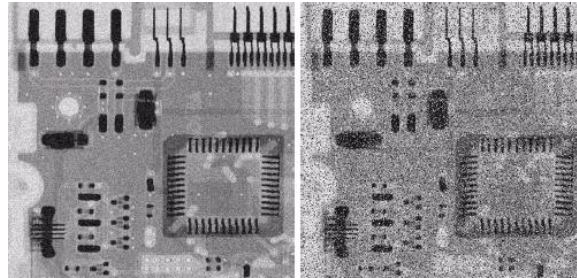
The filter also **removed** some dark points from the borders of the dark objects.

↑
Better job!

The filter also **removed** some white points from the borders of the light objects.

REDUCTION OF NOISE WITH 4 TYPES OF FILTERS

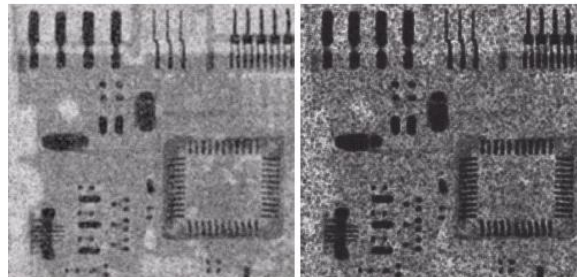
Uniform noise
(mean=0,
variance=800)



Additional salt-and-pepper noise
($P_a=P_b=0.1$)



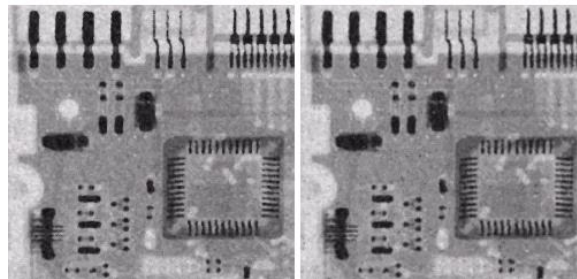
Arithmetic mean
(no good)



Geometric mean
(no good)



Median
(much better)



Alpha-trimmed mean
(better than median)



a b
c d
e f

FIGURE 5.12 (a) Image corrupted by additive uniform noise, (b) Image additionally corrupted by additive salt-and-pepper noise. Image in (b) filtered with a 5×5 : (c) arithmetic mean filter; (d) geometric mean filter; (e) median filter; and (f) alpha-trimmed mean filter with $d = 5$.

ADAPTIVE FILTERS

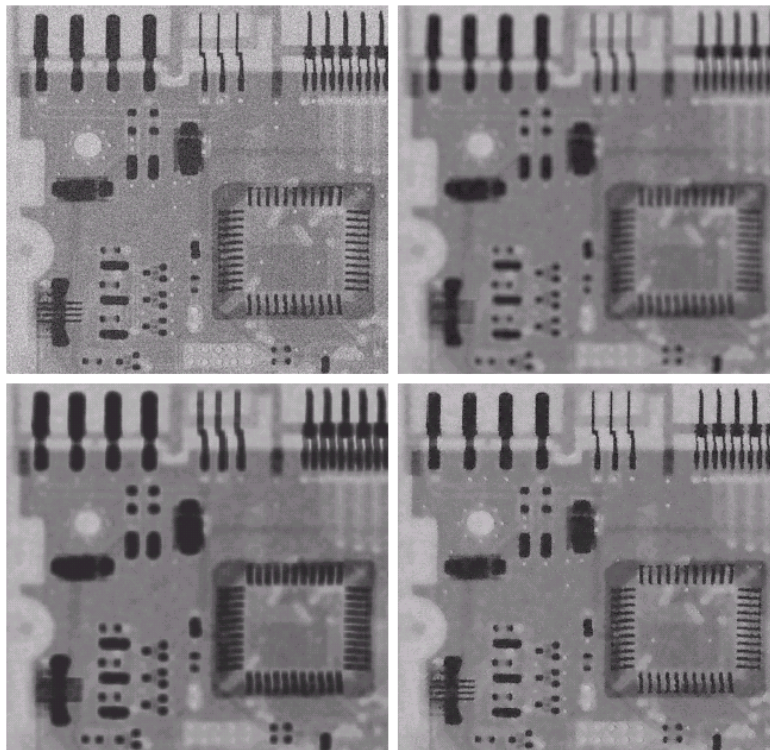
- ❑ The filters discussed so far are applied to an image **independent** of how image characteristics vary from one point to another.
- ❑ 2 simple adaptive filters
 - Adaptive, local noise reduction filter
 - Adaptive median filter
- ❑ Their behavior changes based on statistics of the image inside the filter region.
- ❑ Advantage: superior performance
- ❑ Disadvantage: increase in filter complexity

COMPARISON OF ADAPTIVE FILTER WITH ARITHMETIC AND GEOMETRIC MEAN FILTERS

a b
c d

FIGURE 5.13

(a) Image corrupted by additive Gaussian noise of zero mean and variance 1000.
(b) Result of arithmetic mean filtering.
(c) Result of geometric mean filtering.
(d) Result of adaptive noise reduction filtering. All filters were of size 7×7 .



← **Best results:**

Noise reduction is comparable but the restored image is much sharper!

ADAPTIVE MEDIAN FILTER

$z_{\min}$ = min gray level value in S_{xy}
 $z_{\max}$ = max gray level value in S_{xy}
 z_{med} = median of gray levels in S_{xy}
 z_{xy} = gray level at (x,y)
 $S_{\max}$ = max allowed size of S_{xy}

The filter works in 2 levels:

Level A

$A1 = z_{\text{med}} - z_{\min}$
 $A2 = z_{\text{med}} - z_{\max}$
If $A1 > 0$ and $A2 < 0$, goto level B
Else increase the window size
If window size $\bullet S_{\max}$, repeat level A
Else output z_{xy}

Level B

$B1 = z_{xy} - z_{\min}$
 $B2 = z_{xy} - z_{\max}$
If $B1 > 0$ and $B2 < 0$, output z_{xy}
Else output z_{med}

3 main purposes:

1. To **remove impulsive noise**
2. To **provide smoothing of other noise**
3. To **reduce distortion** (e.g., excessive thinning or thickening of object boundaries)

A SIMPLE EXAMPLE ADAPTIVE MEDIAN FILTER

center
point

10	20	20
20	15	20
20	25	100

$$z_{\min} = 10$$

$$z_{\max} = 100$$

$$z_{\text{med}} = 20$$

$$A1 = 20 - 10 = 10$$

$$A2 = 20 - 100 = -80$$

$A1 > 0$ & $A2 < 0$: $z_{\min} < z_{\text{med}} < z_{\max}$ Hence, z_{med} cannot be an impulse.
Go to level B

$$B1 = 15 - 10 = 5$$

$$B2 = 15 - 100 = -85$$

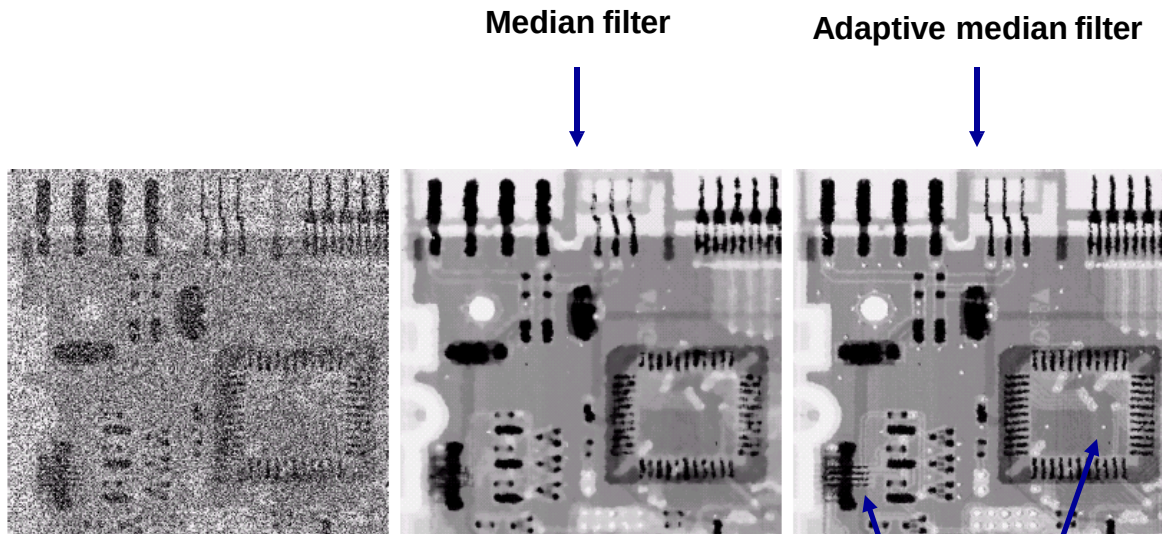
$B1 > 0$ & $B2 < 0$: $z_{\min} < z_{xy} < z_{\max}$ Hence, z_{xy} cannot be an impulse.

Output $z_{xy} = 15$ ← these intermediate-level points are not changed

$(B1 > 0 \text{ \& } B2 < 0)$ is false: $z_{xy} = z_{\min}$ or $z_{xy} = z_{\max}$

Output $z_{\text{med}} = 20$

COMPARISON OF MEDIAN AND ADAPTIVE MEAN FILTERS



a b c

FIGURE 5.14 (a) Image corrupted by salt-and-pepper noise with probabilities $P_a = P_b = 0.25$. (b) Result of filtering with a 7×7 median filter. (c) Result of adaptive median filtering with $S_{\max} = 7$.

Noise reduction is comparable but the filter preserved sharpness!

FREQUENCY DOMAIN FILTERING FOR PERIODIC NOISE

- ❑ **Bandreject** filters: remove or attenuate a band of frequencies about the origin of the Fourier Transform.
- ❑ **Bandpass** filters: pass or strengthen a band of frequencies about the origin of the Fourier Transform .
- ❑ **Notch** filters: reject or pass frequencies in predefined neighborhoods about a center frequency.

BANDREJECT FILTERS

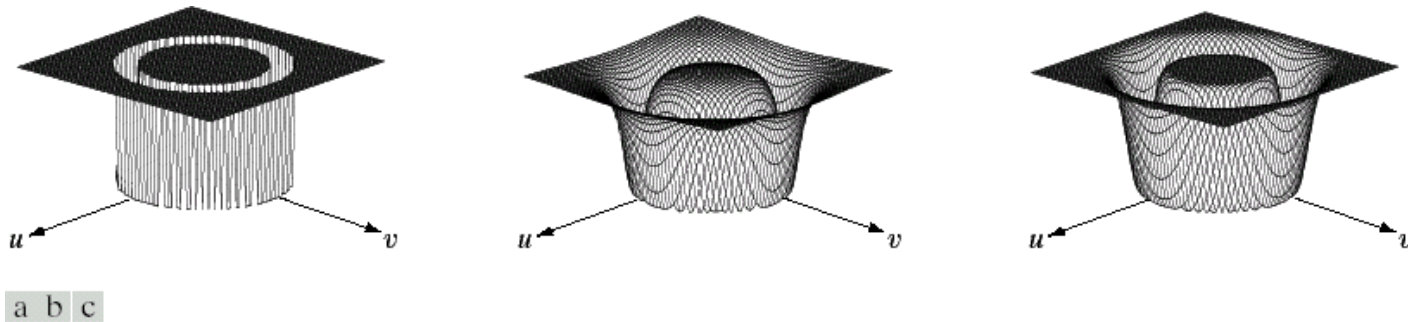


FIGURE 5.15 From left to right, perspective plots of ideal, Butterworth (of order 1), and Gaussian bandreject filters.

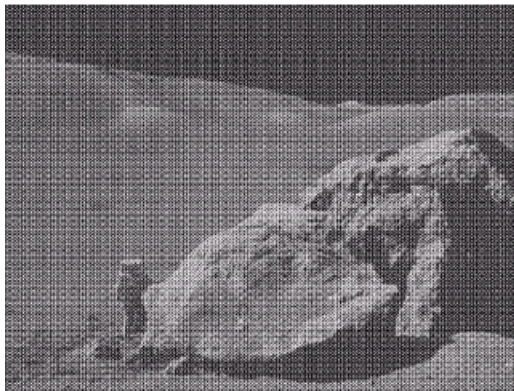
$$H(u, v) = \begin{cases} 1 & \text{if } D(u, v) < D_0 - \frac{W}{2} \\ 0 & \text{if } D_0 - \frac{W}{2} \leq D(u, v) \leq D_0 + \frac{W}{2} \\ 1 & \text{if } D(u, v) > D_0 + \frac{W}{2} \end{cases}$$

$$H(u, v) = \frac{1}{1 + \left[\frac{D(u, v)W}{D^2(u, v) - D_0^2} \right]^{2n}}$$

$$H(u, v) = 1 - e^{-\frac{1}{2} \left[\frac{D^2(u, v) - D_0^2}{D(u, v)W} \right]^2}$$

A **principal** application: noise removal in situations where the general location of the noise components in the frequency domain is approximately known.

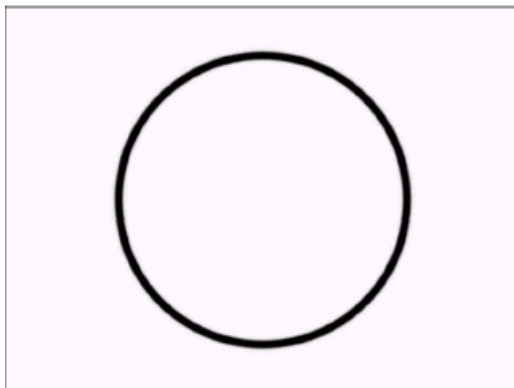
APPLICATION OF A BANDREJECT FILTER



a	b
c	d

FIGURE 5.16

(a) Image corrupted by sinusoidal noise. (b) Spectrum of (a). (c) Butterworth bandreject filter (white represents 1). (d) Result of filtering. (Original image courtesy of NASA.)



Restoration is evident!

Note that it would not be possible to get equivalent results by spatial domain filtering using **small** convolution masks!

BANDPASS FILTERS

$$H_{bp}(u, v) = 1 - H_{br}(u, v)$$



Performs the
opposite operation
of a bandreject
filter.

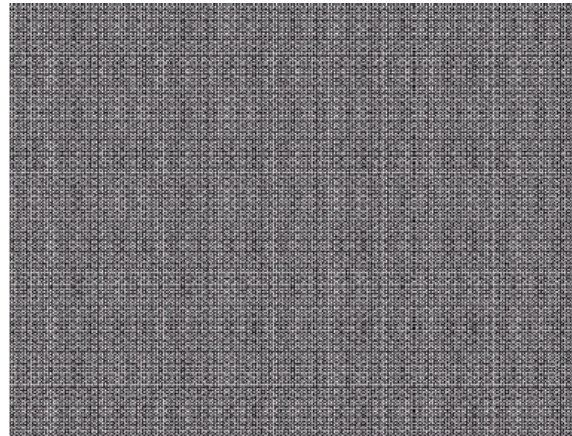
Performing straight bandpass filtering on an image is not a common procedure because it generally removes too much image detail.

Bandpass filtering is quite useful in isolating the effect on an image of selected frequency bands.

APPLICATION OF A BANDPASS FILTER

FIGURE 5.17
Noise pattern of
the image in
Fig. 5.16(a)
obtained by
bandpass filtering.

Most image detail is lost
but the remaining
information is very useful
as it shows a noise
pattern that is close to
that of the noise that
corrupted the image!



Generated by:

- using the bandpass filter corresponding to the bandreject filter in the previous example
- taking the inverse transform

NOTCH FILTERS

- ❑ Notch filters must appear in **symmetric pairs** about the origin in order to obtain meaningful results.
- ❑ The notch filter located at the origin is an **exception**.
- ❑ The **# of pairs** of notch filters is arbitrary.
- ❑ The **shape of the notch areas** is also arbitrary.
- ❑ Two classes
 - Notch **reject** filters
 - Notch **pass** filters
- ❑ Notch reject filters
 - Ideal notch reject filters
 - Butterworth notch reject filters
 - Gaussian notch reject filters
- ❑ Notch pass filters
 - Ideal notch pass filters
 - Butterworth notch pass filters
 - Gaussian notch pass filters

NOTCH REJECT FILTERS

$$H(u, v) = \begin{cases} 0 & D_1(u, v) \leq D_0 \text{ or } D_2(u, v) \leq D_0 \\ 1 & \text{otherwise} \end{cases}$$

$$H(u, v) = \frac{1}{1 + \left[\frac{D_0^2}{D_1(u, v)D_2(u, v)} \right]^n}$$

$$H(u, v) = 1 - e^{-\frac{1}{2} \left[\frac{D_1(u, v)D_2(u, v)}{D_0^2} \right]}$$

Note that these 3 filters become **highpass** filters if $u_0=v_0=0$.

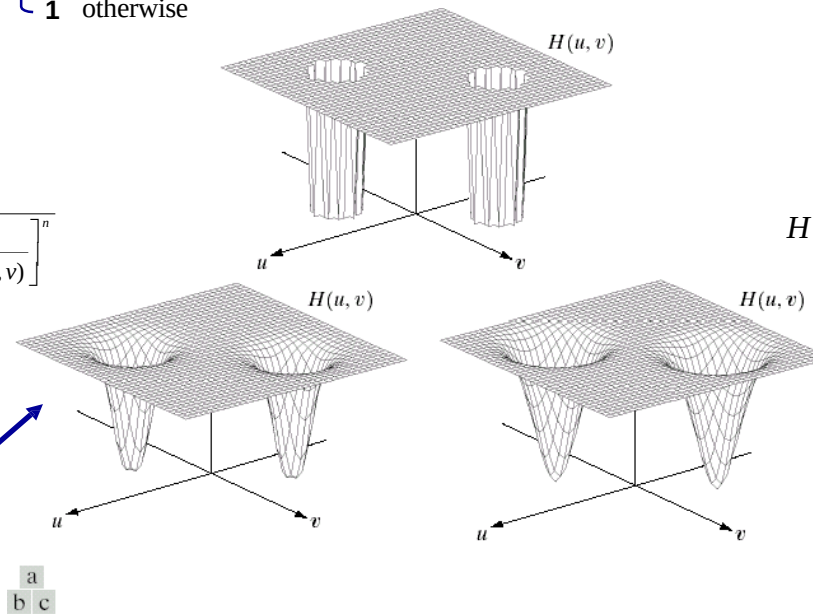


FIGURE 5.18 Perspective plots of (a) ideal, (b) Butterworth (of order 2), and (c) Gaussian notch (reject) filters.

$$D_1(u, v) = [(u - M/2 - u_0)^2 + (v - N/2 - v_0)^2]^{1/2}$$

$$D_2(u, v) = [(u - M/2 + u_0)^2 + (v - N/2 + v_0)^2]^{1/2}$$

NOTCH PASS FILTERS

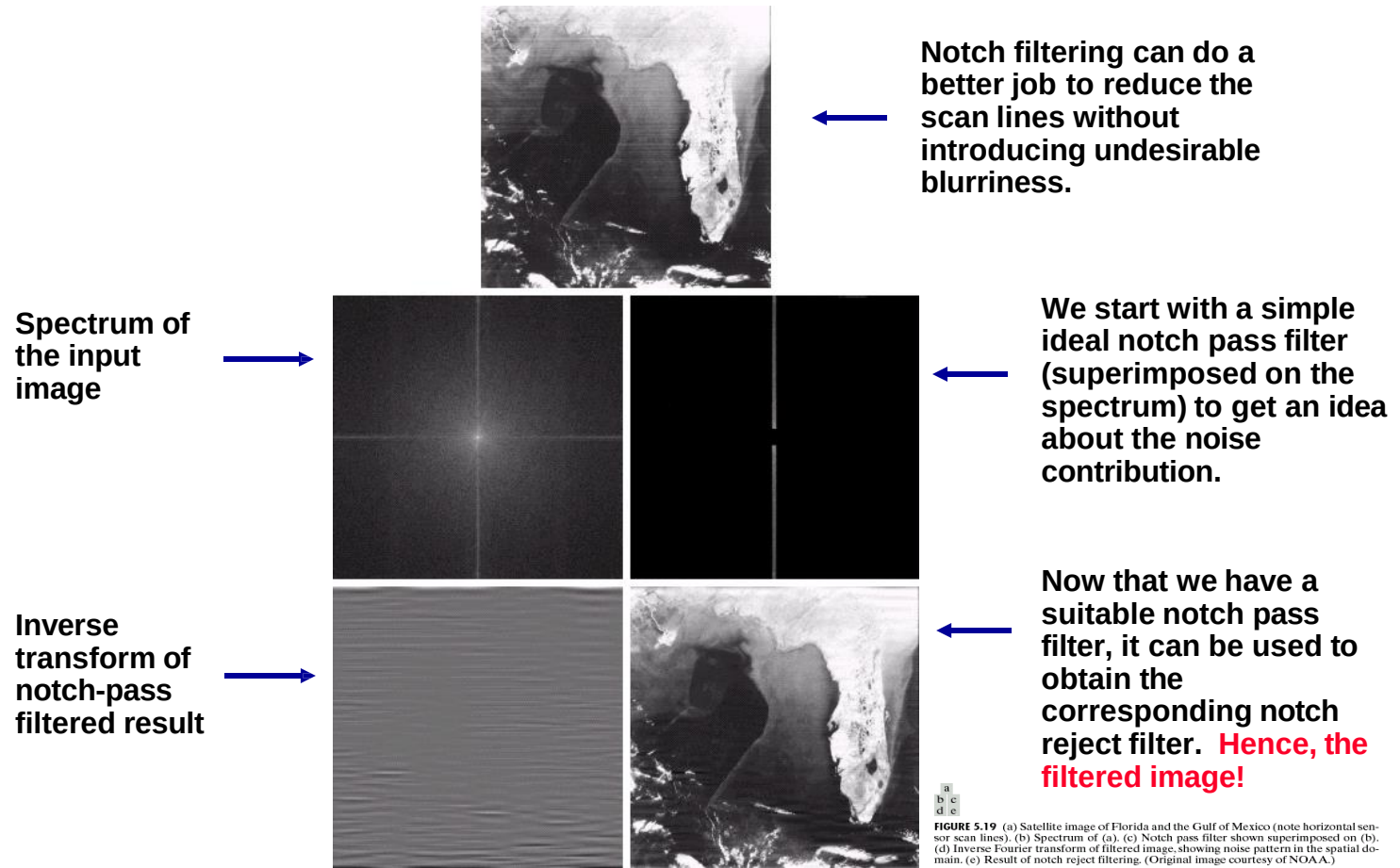
$$H_{np}(u, v) = 1 - H_{nr}(u, v)$$



Performs the
opposite operation
of a notch reject
filter.

Note that the 3 filters become **lowpass** filters if $u_0 = v_0 = 0$.

APPLICATION OF A NOTCH PASS FILTER



LINEAR, POSITION-INVARIANT DEGRADATIONS

- ❑ **Linear, position-invariant** techniques
 - Many types of degradations can be approximated by linear, position-invariant processes.
 - Tools of linear system theory become available.
- ❑ **Nonlinear, position-dependent** techniques
 - More general and usually more accurate
 - Very difficult to solve or no known solutions
- ❑ **Position-invariant operator**
 - The response at any point in the image depends only on the value of the input at that point, not on its position.
- ❑ **If the impulse response of a linear system is known, we can compute the response to any input f !**

DEGRADATION FUNCTION ESTIMATION

- ❑ Degradations are modeled as being the result of convolution.
- ❑ Restoration seeks to find filters that apply the process in reverse.
- ❑ Linear image restoration: **image deconvolution**
- ❑ Restoration filters: **deconvolution filters**
- ❑ 3 principal ways to estimate the degradation function
 - Observation
 - Experimentation
 - Mathematical modeling
- ❑ Image restoration using a degradation function
 - **Blind** deconvolution
 - The true degradation function is seldom known completely.

ESTIMATION BY IMAGE OBSERVATION

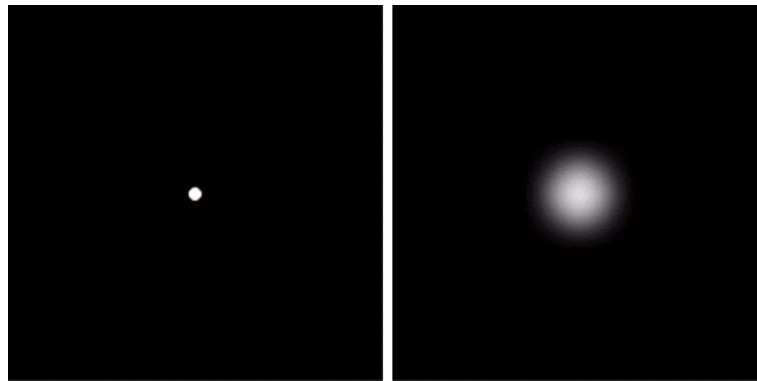
- ❑ Suppose we are given a degraded image without any knowledge about the degradation function H .
- ❑ One way to estimate the function is to gather information from the image itself.
- ❑ Estimation of the degradation function in a blurred image
 - Look for areas of strong signal content (to reduce the effect of noise).
 - Choose a small section of the image with simple structures (e.g., part of an object and the background).
 - Construct an unblurred image of the same size and characteristics of the observed subimage.

$$\left. \begin{array}{l} g_s(x, y): \text{ observed subimage} \\ \hat{f}_s(x, y) : \text{ constructed subimage} \end{array} \right\} H_s(u, v) = \frac{G_s(u, v)}{\hat{F}_s(u, v)}$$

- Suppose a plot of $H_s(u, v)$ looks like a Butterworth filter.
- Construct a function $H(u, v)$ on a larger scale but with the same shape.

ESTIMATION BY EXPERIMENTATION

- ❑ If equipment similar to the equipment used to acquire the degraded image, it is possible to obtain an accurate estimation of the degradation.
- ❑ Estimation of the degradation function
 - Use various system settings until the image is degraded as closely as possible to the image we want to restore.
 - Obtain the impulse response of the degradation by imaging an impulse.
 - $H(u,v) = G(u,v)/A$



a b
FIGURE 5.24
Degradation
estimation by
impulse
characterization.
(a) An impulse of
light (shown
magnified).
(b) Imaged
(degraded)
impulse.

ESTIMATION BY MODELING

- ❑ Provides insight into the restoration problem.
- ❑ In some cases, models take into account environmental conditions.

a b
c d

FIGURE 5.25
Illustration of the
atmospheric
turbulence model.
(a) Negligible
turbulence.
(b) Severe
turbulence,
 $k = 0.0025$.
(c) Mild
turbulence,
 $k = 0.001$.
(d) Low
turbulence,
 $k = 0.00025$.
(Original image
courtesy of
NASA.)



$k = 0.001$

$k = 0.0025$

$k = 0.00025$

$$H(u, v) = e^{-k(u^2 + v^2)^{5/6}}$$

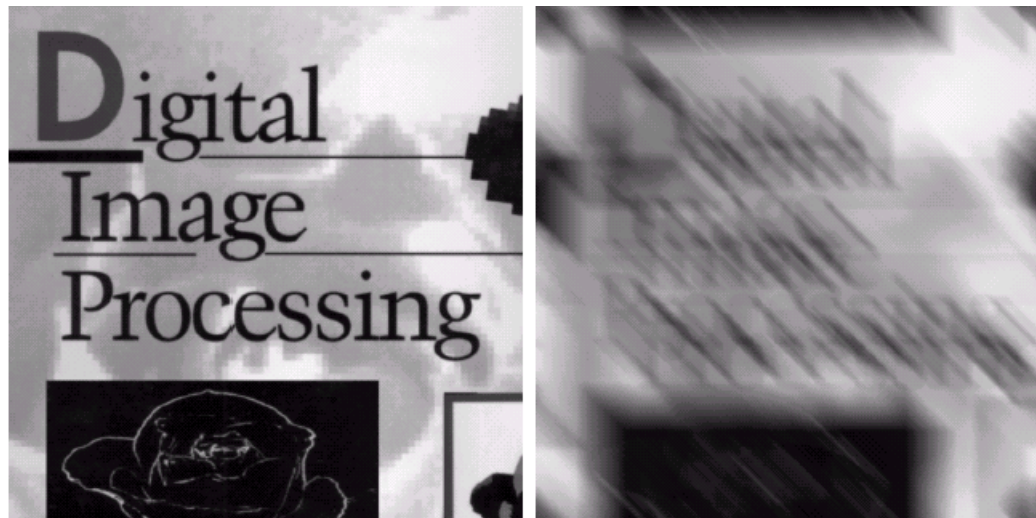
A degradation model
based on the
physical
characteristics of
atmospheric
turbulence.

Sometimes, Gaussian
LPF is used to model
mild, uniform
blurring.

AN EXAMPLE OF MODELING

$$H(u, v) = \frac{T}{\pi(ua + vb)} \sin[\pi(ua + vb)] e^{-j\pi(ua + vb)}$$

Model derived from an image that has been blurred by uniform linear motion between the image and the sensor. ←



a b

FIGURE 5.26 (a) Original image. (b) Result of blurring using the function in Eq. (5.6-11) with $a = b = 0.1$ and $T = 1$.

INVERSE FILTERING

- ❑ **Direct inverse filtering:** The simplest approach to restoration.
- ❑ The following derivation shows that we cannot recover the undegraded image exactly because $N(u,v)$ is a random function whose Fourier Transform is not known!

$$\hat{F}(u, v) = \frac{G(u, v)}{H(u, v)}$$

$$G(u, v) = H(u, v)F(u, v) + N(u, v)$$

$$\Rightarrow \hat{F}(u, v) = F(u, v) + \frac{N(u, v)}{H(u, v)}$$

- ❑ **More bad news:** If the degradation has zero or very small values, the ratio $N(u,v)/H(u,v)$ may dominate the estimate.
 - To circumvent the problem, we can limit the filter frequencies to values near the origin.
 - $H(0,0)$ = Mean of $h(x,y)$: usually the highest value of $H(u,v)$.

AN EXAMPLE OF INVERSE FILTERING

a b
c d

FIGURE 5.27

Restoring Fig. 5.25(b) with Eq. (5.7-1). (a) Result of using the full filter. (b) Result with H cut off outside a radius of 40; (c) outside a radius of 70; and (d) outside a radius of 85.



Degradation function:

$$H(u, v) = e^{-k[(u-M/2)^2 + (v-N/2)^2]^{5/6}}$$

with $k = 0.0025$

The cutoff was implemented by applying a **Butterworth lowpass** function of order 10.

MINIMUM MEAN SQUARE ERROR (WIENER) FILTERING

- ❑ Inverse filtering makes no explicit provision for handling noise.
- ❑ Now we consider both images and noise as random processes.
- ❑ The objective: Find an estimate $\hat{f}$ of the uncorrupted image f such that the mean squared error between them is minimized.
- ❑ Assumptions
 - The image and the noise are uncorrelated.
 - The image or the noise has zero mean.
 - The gray levels in the estimate are a linear function of the levels in the degraded image.
- ❑ The restored image is given by the inverse Fourier transform of $\hat{F}(u, v)$.
- ❑ Noise = zero ➡ **Wiener filter = inverse filter!**

DERIVATION OF THE WIENER FILTER

Minimize $e^2 = E\{(f - \hat{f})^2\}$

$$\Rightarrow \hat{F}(u, v) = \left[\frac{1}{H(u, v)} \frac{|H(u, v)|^2}{|H(u, v)|^2 + S_\eta(u, v) / S_f(u, v)} \right] G(u, v)$$

$H(u, v)$ = transform of the degradation function

$G(u, v)$ = transform of the degraded image

$S_\eta(u, v) = |N(u, v)|^2$ = power spectrum of the noise

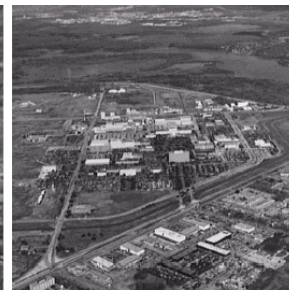
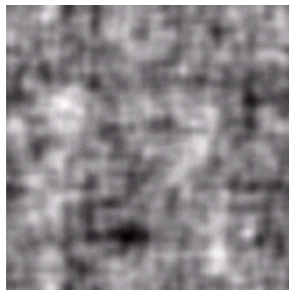
$S_f(u, v) = |F(u, v)|^2$ = power spectrum of the undegraded image

If the 2 power spectrums are not known, use $S_\eta(u, v) / S_f(u, v) = K$

AN EXAMPLE OF WIENER FILTERING



← Degraded input image



K was chosen **interactively** to yield the best possible visual results.

a b c

FIGURE 5.28 Comparison of inverse- and Wiener filtering. (a) Result of full inverse filtering of Fig. 5.25(b). (b) Radially limited inverse filter result. (c) Wiener filter result.

Previous results for comparison

AN EXAMPLE OF WIENER FILTERING

Blurred image with
additive Gaussian
noise (mean=0,
variance=650)



Noise variance is
reduced by one
order of magnitude



Noise variance is
reduced by five
orders of magnitude



Wiener filter with

$$H(u, v) = e^{-k[(u-M/2)^2 + (v-N/2)^2]^{5/6}}$$

where $k = 0.0025$.

K was chosen interactively.

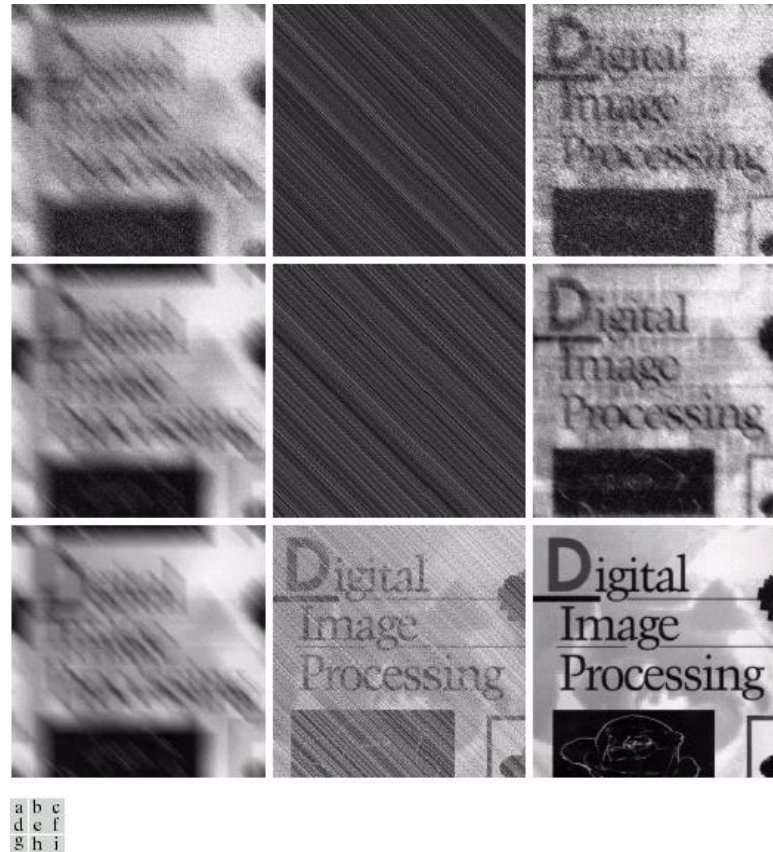


FIGURE 5.29 (a) Image corrupted by motion blur and additive noise. (b) Result of inverse filtering. (c) Result of Wiener filtering. (d)–(f) Same sequence, but with noise variance one order of magnitude less (g)–(i) Same sequence, but noise variance reduced by five orders of magnitude from (a). Note in (h) how the deblurred image is quite visible through a “curtain” of noise.



Thanks